



APPROVED

By John Ozoga III at 10:01 am, Apr 30, 2024

March 28, 2024

Michigan Department of Environment, Great Lakes and Energy
Materials Management Division
Cadillac District Office
120 W. Chapin Street
Cadillac, Michigan 49601
Submitted via email

Attn: John Ozoga, III, District Supervisor/Engineer
ozogaj@michigan.gov

Subject: **PART 115 Surface Monitoring Plan**
Wexford County Landfill
Manton, Michigan

Dear Mr. Ozoga:

Attached please find the above referenced PART 115 Surface Monitoring Plan for the Wexford County Landfill. The Plan was prepared by Tetra Tech in response to the March 22, 2023, "Landfill Gas Amendments Advisory Letter" from the Michigan Department of Environment, Great, Lakes, and Energy (EGLE) to GFL Environmental USA Inc.

Should you have any questions regarding the attached Plan, please contact me at (770) 575-7610.

Sincerely,

A handwritten signature in blue ink, appearing to read 'Tami Craig'.

Tami Craig
Regional Landfill Gas Program Manager

attached: PART 115 Surface Monitoring Plan

cc: C. Gee, cgee@gflenv.com
C. Oleniacz, doleniacz@eilllc.com
T.Unseld, unseldt@michigan.com

**PART 115 Surface Monitoring Plan
For
Wexford County Landfill**

Manton, Michigan



March 28, 2024

Prepared By:

Tetra Tech
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1.0 Purpose

This document serves as the Part 115 Surface Monitoring Plan (SMP) for Wexford County Landfill (“the Facility”) in accordance with the requirements of Part 115, Subpart 2, Section 11512. The effective date of this statute was March 29, 2023.

2.0 Applicability

Section 11512d(6) requires that surface emissions monitoring (SEM) be conducted in accordance with a surface monitoring design plan approved by the department. However, no date or instructions for submittal of this plan was included in the amended statute.

On March 22, 2023 EGLE sent the Facility a “Landfill Gas Amendments Advisory Letter” with implementation schedules for various categories of landfill facilities. For sites with an active collection system, but not yet subject to the NESHAP, such as Wexford County Landfill, a surface monitoring plan (SMP) was required for submittal one year from the effective date; i.e. by March 29, 2024.

Note that the Facility would not normally be subject to SEM requirements under federal air programs since these only apply to those landfills which have a non-methane organic compound (NMOC) emissions rate of greater than 34 Mg/year. The Facility has demonstrated via Tier 2 testing conducted in accordance with 40 CFR 62 Subpart OOO (and previously 40 CFR 60 Subpart WWW) that NMOC emissions rates are below federal regulatory requirements for installation of controls and subsequent compliance monitoring. However, the State of Michigan’s revised solid waste regulations now require this monitoring regardless of facility status under the federal air regulations.

3.0 Surface Emissions Monitoring Design Plan – Section 11512d(6)

Section 11512h(2) states that the owner or operator of a Type II landfill with an active landfill gas collection and control system shall begin surface emissions scans within one year after the effective date of the amendatory act that added the new section.

Note that EGLE’s March 22, 2023 Landfill Gas Amendments Advisory Letter later clarified that monitoring should commence after approval of the SMP. This Surface Emissions Monitoring Design Plan presented herein establishes protocol for this monitoring consistent with Section 11512d requirements. The procedures for conducting surface emissions testing will be in accordance with EPA Method 21.

The purpose of the monitoring is to establish that the methane concentration is 500 parts per million (ppm) or less above background at the surface of the landfill. Section 11512f(c) states that design plans for an active gas collection and control system shall be submitted within one year after detecting surface methane emissions at concentrations exceeding 500 ppm above background that cannot be corrected within one quarterly monitoring period. Furthermore, construction of the gas system must be completed within 180 days of department approval of the design plan, unless an extension is approved by the department.

3.1 SEM Frequency

Although not explicitly stated in Section 11512d(5), the frequency of SEM monitoring is presumed to be quarterly for active sites and annually for portions of sites that have been final closed under the Part 115 rules. The definition of “final closed” is that the area has had certified final cover placed. Any portions of the site that were final closed under prior solid waste regulations (such as Act 641 or Act 87) are NOT subject to SEM monitoring at all, and these areas (if any) will be shown as exempt on the surface scan map.

For the areas of the site that have been final closed under Part 115, once three quarters of exceedance-free SEM monitoring have been demonstrated, the facility will revert to an annual schedule in these areas. Should an exceedance be detected in a closed area during an annual SEM event, SEM monitoring will revert back to quarterly until three subsequent quarters of SEM monitoring without an exceedance are demonstrated. At that point, the closed area can revert back to an annual SEM monitoring frequency. SEM monitoring will be discontinued altogether once active waste receipt ceases and the facility is permanently closed.

At the Facility, there are separate landfill areas that have been certified as closed. The facility will conduct three initial rounds of quarterly monitoring at this closed landfill area. Should no exceedances be detected during these first three quarterly events, the facility will then revert to an annual monitoring schedule for the closed area. This is in accordance with the following provision of the federal air regulations:

§62.16722(f) Any closed landfill that has no monitored exceedances of the operational standard in three consecutive quarterly monitoring periods may skip to annual monitoring. Any reading of 500 ppm or more above background detected during the annual monitoring returns the frequency for the landfill to quarterly monitoring.

3.2 Initiation of SEM in New Waste Areas – Section 11512d(5)

Not later than 180 days after initial waste receipt in a portion of a landfill, surface emissions monitoring for methane will be commenced at all of the following locations:

- (a) Where visual observations, such as of distressed vegetation or cracks or seeps in the cover, indicate elevated concentrations of landfill gas.*
- (b) At each penetration of daily, interim, or final landfill cover.*
- (c) Around the perimeter of the active gas collection and control system.*
- (d) Along a pattern that traverses the landfill at no more than 30-meter intervals, unless the owner or operator establishes an alternative traversing pattern that is approved by the department after a site-specific demonstration. NOTE: The facility is not proposing to establish an alternative traversing pattern.*

The definition of cover penetration that will be used by the site for SEM monitoring is the definition provided in the Landfill NESHAP (40 CFR 63 Subpart AAAA - §63.1990):

Cover penetration means a wellhead, a part of a landfill gas collection or operations system, and/or any other object that completely passes through the landfill cover. The landfill cover includes that portion which covers the waste, as well as the portion which borders the waste

extended to the point where it is sealed with the landfill liner or the surrounding land mass. Examples of what is not a penetration for purposes of this subpart include but are not limited to: Survey stakes, fencing including litter fences, flags, signs, utility posts, and trees so long as these items do not pass through the landfill cover.

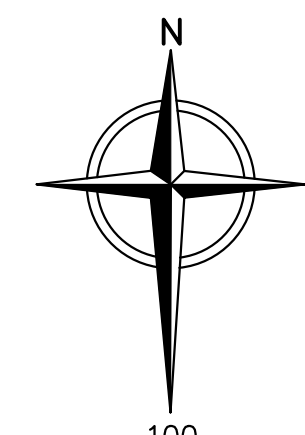
3.3 Monitoring Route and Exclusions – Section 11512d(6)

The proposed surface monitoring topographical route map for the Facility at final grade conditions (i.e. closure) is provided in Appendix A. The initial monitoring route proposed to be followed in 2024 is provided in Appendix B.

Actual routes followed will be documented in each quarterly report and kept in the site's operating record. The path followed while the landfill is actively filling will vary depending on the footprint of in-place waste at the time of the survey. However, the route will meet the requirements of Section 11512d(5) with respect to route pattern, spacing, cover penetrations, in-place waste perimeter and visual observation results of potential elevated landfill gas concentrations. Areas excluded from each survey due to dangerous conditions will be documented on the route map for each quarterly survey. Dangerous conditions will be determined by the monitoring technician at the time of each SEM event. Dangerous conditions are defined as those conditions that may present a substantial risk of harm or potential loss of life to the individual conducting the monitoring, and can include steep interim waste slopes, icy or slippery conditions, truck traffic from construction or cover placement, the active working face, etc.

Appendix A

Surface Monitoring Plan Route Map – Final Grades



0 100 200

LEGEND

- PROPERTY BOUNDARY
- EXISTING SOLID WASTE BOUNDARY
- EXISTING PHASE BOUNDARY
- PROPOSED 10' CONTOUR
- PROPOSED 2' CONTOUR
- EXISTING LANDFILL GAS HEADER
- EXISTING LANDFILL GAS HORIZONTAL COLLECTOR
- EXISTING FORCEMAIN
- EXISTING 3" FORCEMAIN
- EXISTING 3"x6" DUAL CONTAINED FORCEMAIN LINE
- EXISTING 4"x8" DUAL CONTAINED FORCEMAIN LINE
- EXISTING AIRLINE
- EXISTING 2" AIRLINE
- EXISTING LFG EXTRACTION WELL
- EXISTING LFG EXTRACTION WELL WITH CAISSON
- EXISTING REMOTE WELLHEAD
- EXISTING HEADER ACCESS RISER
- EXISTING ISOLATION VALVE
- EXISTING BLIND FLANGE
- EXISTING FLANGE CONNECTION
- EXISTING REDUCER FITTING
- EXISTING CONDENSATE DRIPLEG
- EXISTING MANHOLE
- EXISTING FORCEMAIN ISOLATION VALVE
- EXISTING AIRLINE ISOLATION VALVE
- EXISTING FORCEMAIN CLEANOUT
- EXISTING FLANGE CONNECTION
- EXISTING CONDENSATE SUMP
- EXISTING FINAL COVER AREA
- PROPOSED SURFACE EMISSIONS MONITORING PATHWAY

NOTES:

- GRADES SHOWN REFLECT ANTICIPATED FINAL DEVELOPMENT ELEVATIONS, PROVIDED BY WEAVER CONSULTANTS GROUP, INC. IN FILE TITLED 'WC TOW GRADES'.
- SURFACE EMISSIONS MONITORING FOR METHANE TO BE PERFORMED IN ACCORDANCE WITH PART 115 SECTION 11512d, DESCRIBED IN SECTION 3.0 OF THE PART 115 PLAN FOR WEXFORD COUNTY LANDFILL.
- SURFACE EMISSIONS MONITORING 'SERPENTINE' PATHWAY REPRESENTS FINAL GRADE CONDITIONS. INTERIM PATHWAYS WILL BE MONITORED BASED UPON THE SEQUENCE AND EXTENT OF WASTE PLACEMENT AND GCS DEVELOPMENT.
- TYPICAL SPACING FOR SERPENTINE PATHWAY WILL BE NO MORE THAN 30 METERS (100 FT), SOME LOCAL VARIATIONS MAY OCCUR TO ACCOMMODATE LANDFILL PHASING AND FIELD CONDITIONS AT THE TIME OF MONITORING.
- IN ADDITION TO SURFACE EMISSIONS MONITORING FOR METHANE ALONG THE SERPENTINE PATHWAY, SURFACE EMISSIONS MONITORING FOR METHANE SHALL ALSO BE CONDUCTED AT EACH COVER PENETRATION AND WHERE VISUAL OBSERVATION INDICATES ELEVATED CONCENTRATION OF LANDFILL GAS (SPECIFIED IN SECTION 3.2 OF THE PART 115 PLAN FOR WEXFORD COUNTY LANDFILL).
- THE FACILITY MAY EXCLUDE DANGEROUS AREAS, SUCH AS ROADS, THE ACTIVE FILL AREA, TRUCK TRAFFIC AREAS, AREAS WITH SNOW OR ICE COVER AND SLOPES STEEPER THAN OR EQUAL TO 3:1 FROM SURFACE TESTING.

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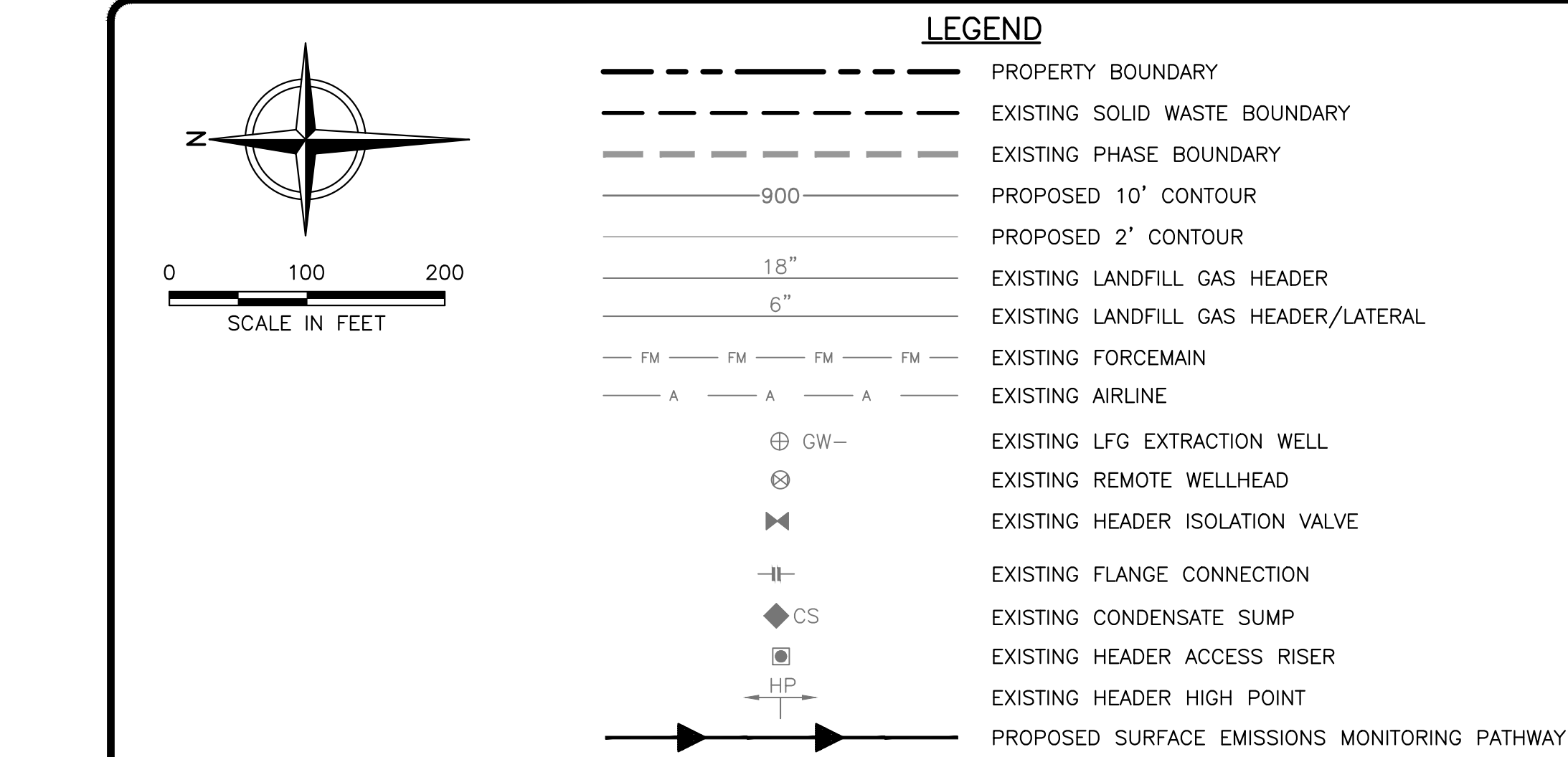
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WEXFORD COUNTY LANDFILL
WEXFORD COUNTY, MICHIGAN

**2024 LANDFILL GAS MONITORING AND MANAGEMENT PLAN
SEM PLAN - FINAL DEVELOPMENT**

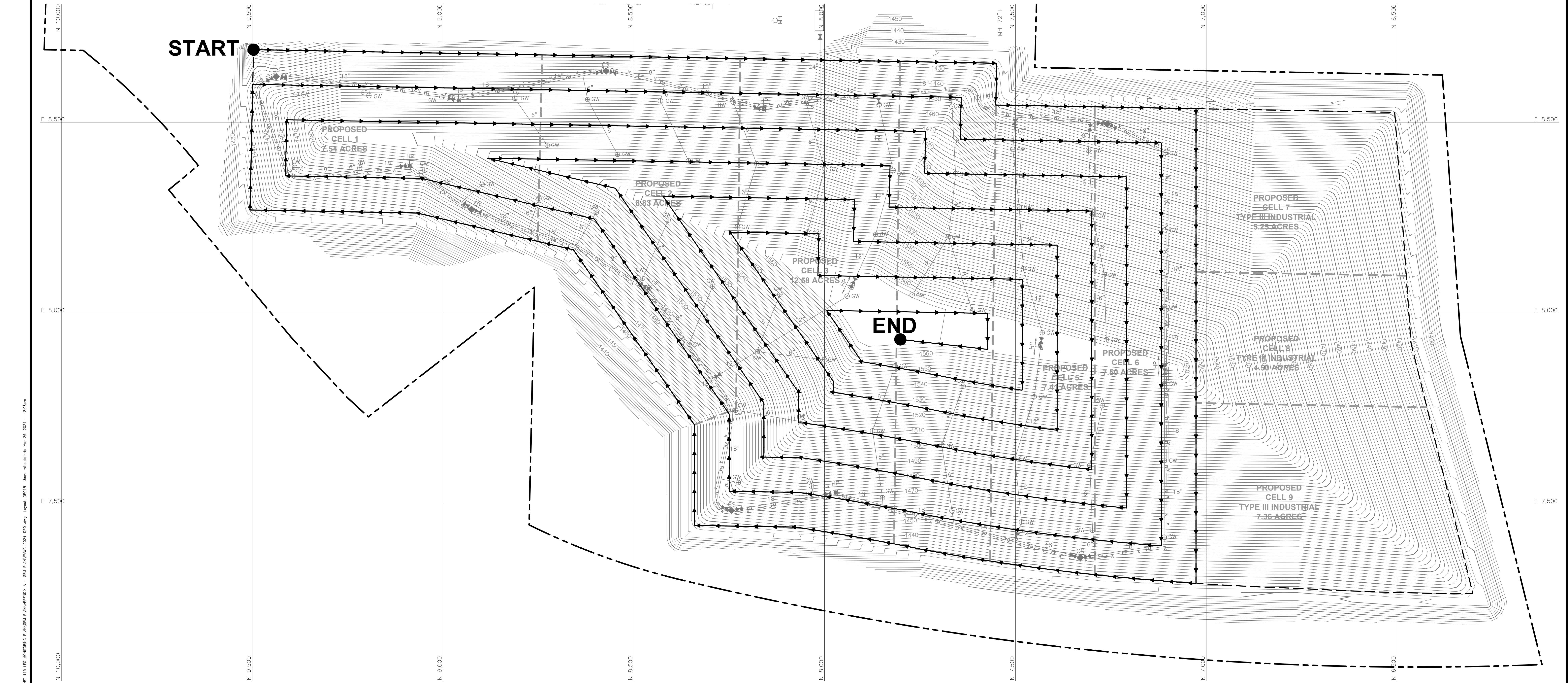
SHEET NO.
SEM 1A

PROJECT NO.
209-4241336

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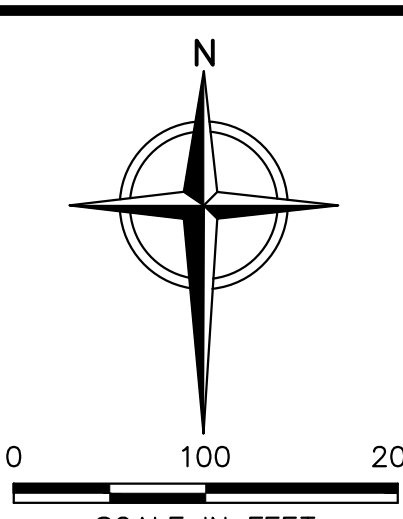
**2024 LANDFILL GAS MONITORING AND MANAGEMENT PLAN
SEM PLAN - FINAL DEVELOPMENT**

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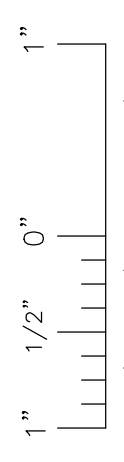
Appendix B

Surface Monitoring Plan Map – Current Grades in 2024



LEGEND	
	PROPERTY BOUNDARY
	EXISTING SOLID WASTE BOUNDARY
	EXISTING PHASE BOUNDARY
	EXISTING 10' CONTOUR
	EXISTING 2' CONTOUR
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	EXISTING FORCEMAIN ISOLATION VALVE
	EXISTING AIRLINE ISOLATION VALVE
	EXISTING FORCEMAIN CLEANOUT
	EXISTING BLIND FLANGE
	EXISTING FLANGE CONNECTION
	EXISTING REDUCER FITTING
	EXISTING CONDENSATE SUMP
	EXISTING HORIZONTAL COLLECTOR ROCK SUMP
	EXISTING PUMP
	EXISTING ANTI-SEEP COLLAR
	EXISTING HIGH POINT
	EXISTING LOW POINT
	EXISTING CONDENSATE DRIPLEG
	EXISTING MANHOLE
	EXISTING FORCEMAIN ISOLATION VALVE
	EXISTING AIRLINE ISOLATION VALVE
	EXISTING FORCEMAIN CLEANOUT
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**2024 LANDFILL GAS MONITORING AND MANAGEMENT PLAN
SEM PLAN - EXISTING CONDITIONS**

SHEET NO.
SEM 2

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